

Sodium percarbonate

Sodium percarbonate, or **sodium carbonate peroxide** is a chemical substance with formula Na2H3CO6. It is an adduct of sodium carbonate ("soda ash" or "washing soda") and hydrogen peroxide (that is, a perhydrate) whose formula is more properly written as 2 Na2CO3 · 3 H2O2. It is a colorless, crystalline, hygroscopic and water-soluble solid.^[3] It is sometimes abbreviated as **SPC**. It contains 32.5% by weight of hydrogen peroxide.

The product is used in some eco-friendly bleaches and other cleaning products.^[3]

History

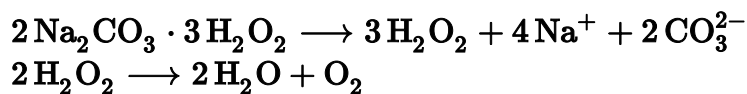
Sodium percarbonate was first prepared in 1899 by Ukrainian chemist Sebastian Moiseevich Tanatar (7 October 1849 – 30 November 1917).^[4]

Structure

At room temperature, solid sodium percarbonate has the orthorhombic crystal structure, with the *Cmca* crystallographic space group. The structure changes to *Pbca* as the crystals are cooled below about −30 °C.^[1]

Chemistry

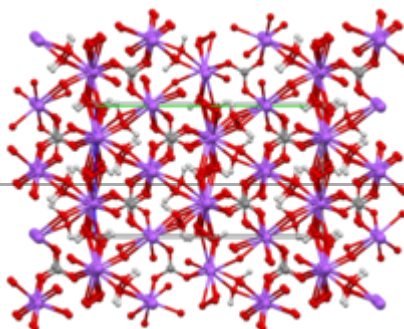
Dissolved in water, sodium percarbonate yields a mixture of hydrogen peroxide (which eventually decomposes to water and oxygen), sodium cations Na+, and carbonate CO3^{2−}.^{[3][5]}



Production

Sodium percarbonate is produced industrially by crystallization of a solution of sodium carbonate and hydrogen peroxide, with proper control of the pH and concentrations.^{[6][1][7]} This is also a convenient laboratory method.

Sodium percarbonate



Crystal structure at 100 K ^[1]

Names

IUPAC name

sodium carbonate—hydrogen peroxide (2/3)

Other names

Sodium carbonate peroxide,^[2]
sodium carbonate
sesquiperhydrate, PCS, SPC,
solid hydrogen peroxide,
Sodium carbonate hydrogen
peroxide, sodium carbonate
peroxyhydrate

Identifiers

CAS Number

15630-89-4 (https://commonchemistry.org/detail?cas_rn=15630-89-4) ✓

3D model (JSmol)

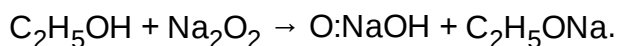
Interactive image (<https://chemapps.stolaf.edu/jmol/jmol.php?model=%5BNa%2B%5D.%5BO-%5DC%28%3DO%29O>)

ChemSpider

140471 (<https://www.chemspider.com/C>)

Alternatively, dry sodium carbonate may be treated directly with concentrated hydrogen peroxide solution.^[8]

It may also be formed from a process starting from sodium peroxide; when absolute ethyl alcohol reacts with sodium peroxide at 0 °C, a perhydroxide is produced.



Carbon dioxide converts it into sodium hydrogen percarbonate.

World production capacity of this compound was estimated at several hundred thousand tons for 2004.^[9]

Uses

As an oxidizing agent, sodium percarbonate is an ingredient in a number of home and laundry cleaning products, including non-chlorine bleach products such as Oxyper, OxiClean, Tide laundry detergent,^[3] and Vanish.^[5]

Many commercial products mix a percentage of sodium percarbonate with sodium carbonate. The average "Oxy" product in the supermarket contains 35–40% sodium percarbonate with about 5% active oxygen when titrated.

Sodium percarbonate is also used as a cleaning agent in homebrewing.^[10]

Sodium percarbonate can be used in organic synthesis as a convenient source of anhydrous H_2O_2 , in particular in solvents that cannot dissolve the carbonate but can leach the H_2O_2 out of it.^[11] A method for generating trifluoroperacetic acid *in situ* for use in Baeyer–Villiger oxidations from sodium percarbonate and trifluoroacetic anhydride has been reported; it provides a convenient and cheap approach to this reagent without the need to obtain highly concentrated hydrogen peroxide.^{[12][13]}

References

1. R. G. Pritchard & E. Islam (2003). "Sodium percarbonate between 293 and 100 K". *Acta Crystallographica Section B*. **B59** (5): 596–605. doi:10.1107/S0108768103012291 (https://doi.org/10.1107/S0108768103012291). PMID 14586079 (https://pubmed.ncbi.nlm.nih.gov/14586079).
2. "Substance Name: Sodium carbonate peroxide" (https://chem.nlm.nih.gov/chemidplus/rn/15630-89-4). Retrieved 2021-09-09.

	hemical-Structure.140471.html) ✓
ECHA InfoCard	100.036.082 (https://echa.europa.eu/substance-information/-/substanceinfo/100.036.082)
EC Number	239-707-6
PubChem CID	159762 (https://pubchem.ncbi.nlm.nih.gov/compound/159762)
RTECS number	FG0750000
UNII	Z7G82NV92P (https://precision.fda.gov/uniisearch/srs/unii/Z7G82NV92P) ✓
CompTox Dashboard (EPA)	DTXSID3029736 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID3029736)
InChI	InChI=1S/CH2O4.Na/c2-1(3)5-4;/h4H,(H,2,3);/q;+1/p-1 ✓ Key: MWNQXXOSWHCCOZ-UHFFFAOYSA-M ✓ InChI=1S/CH2O4.Na/c2-1(3)5-4;/h4H,(H,2,3);/q;+1/p-1 Key: MWNQXXOSWHCCOZ-REWHXWOFAO Key: MWNQXXOSWHCCOZ-UHFFFAOYSA-M
SMILES	[Na+].[O-]C(=O)OO
Properties	
Chemical formula	Na ₂ CO ₃ ·1.5 H ₂ O ₂
Molar mass	156.982 g/mol
Appearance	White solid
Solubility in water	150 g/l

3. Craig W. Jones (1999). *Applications of hydrogen peroxide and its derivatives*. Royal Society of Chemistry. ISBN 0-85404-536-8.
4. Tanatar, S. (1899). "Percarbonate" (<https://babel.hathitrust.org/cgi/pt?id=hvd.cl1i1w&view=1up&seq=198&skin=2021>). *Berichte der Deutschen Chemischen Gesellschaft zu Berlin* (in German). **32** (2): 1544–1546. doi:10.1002/cber.18990320233 (<https://doi.org/10.1002%2Fcber.18990320233>).
5. "Oxygen-based bleaches (<http://www.chemistryinyourcupboard.org/vanish/4>) Archived (<https://web.archive.org/web/20120124185550/http://www.chemistryinyourcupboard.org/vanish/4>) 2012-01-24 at the Wayback Machine", The Royal Society of Chemistry, and Reckitt Benckiser (the manufacturers of Vanish).
6. J. M. Adams and R. G. Pritchard (1977): "The crystal structure of sodium percarbonate: an unusual layered solid". *Acta Crystallographica Section B*, volume B33, issue 12, pages 3650–3653. doi:10.1107/S0567740877011790 (<https://doi.org/10.1107%2FS0567740877011790>)
7. Alun P. James, Graham R. Horne, Richard Roesler, and others (1997): "Process for producing sodium percarbonate (<https://patents.google.com/patent/US6231828>)". US Patent US6231828B1, priority date 1997-03-26.
8. Sang Ryul Kim, Chong Yun Kwag, Hwan Kee Heo, Jong-Pill Lee (1996): "Process for manufacturing granular sodium percarbonate (<https://patents.google.com/patent/US5851420>)". US Patent US5851420A, priority date 1996-02-29
9. Harald Jakob, Stefan Leininger, Thomas Lehmann, Sylvia Jacobi, Sven Gutewort. "Peroxo Compounds, Inorganic". *Ullmann's Encyclopedia of Industrial Chemistry*. Weinheim: Wiley-VCH. doi:10.1002/14356007.a19_177.pub2 (https://doi.org/10.1002%2F14356007.a19_177.pub2).
10. "Sodium Percarbonate" (<https://www.morebeer.com/products/sodium-percarbonate.html>). *MoreBeer.com*. Retrieved 26 June 2020.
11. McKillop, A (1995). "Sodium perborate and sodium percarbonate: Cheap, safe and versatile oxidising agents for organic synthesis". *Tetrahedron*. **51** (22): 6145–6166. doi:10.1016/0040-4020(95)00304-Q (<https://doi.org/10.1016%2F0040-4020%2895%2900304-Q>).

Hazards	
Occupational safety and health (OHS/OSH):	
Main hazards	Irritant, oxidizer
Flash point	Non-flammable
Related compounds	
Other anions	Sodium carbonate Sodium bicarbonate
Other cations	Calcium percarbonate Magnesium percarbonate
Related compounds	Sodium perborate Sodium persulfate Sodium perphosphate
Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa).	
Infobox references	

12. Kang, Ho-Jung; Jeong, Hee-Sun (1996). "New Method of Generating Trifluoroperoxyacetic acid for the Baeyer-Villiger Reaction" (http://journal.kcsnet.or.kr/main/j_search/j_abstract_view.htm?code=B960104&cpage=3&qpage=j_search&spage=j_search&journal=B&vol=17&no=1&page=&year1=1990&year2=1999&view=10&qpage=j_search&abstract=). *Bull. Korean Chem. Soc.* **17** (1): 5–6.
13. Caster, Kenneth C.; Rao, A. Somasekar; Mohan, H. Rama; McGrath, Nicholas A.; Brichacek, Matthew (2012). "Trifluoroperacetic Acid". *Encyclopedia of Reagents for Organic Synthesis. e-EROS Encyclopedia of Reagents for Organic Synthesis*. doi:10.1002/047084289X.rt254.pub2 (<https://doi.org/10.1002%2F047084289X.rt254.pub2>). ISBN 978-0471936237.

External links

- Organic Chemistry Portal: Sodium percarbonate (<https://www.organic-chemistry.org/chemicals/oxidations/sodiumpercarbonate.shtml>)
 - Consumer Product Information Database: Sodium percarbonate (<https://www.whatsinproducts.com/chemicals/view/1/3259/003313-92-6/Sodium%20percarbonate>)
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